

Serie 13

INVERTIBILITY

1. Inverses. For each of the following matrices, determine whether or not it is invertible, and if it is compute its inverse.

(a) $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$

(d) $\begin{pmatrix} 1 & 2 & -3 & 1 \\ -1 & 3 & -3 & -2 \\ 2 & 0 & 1 & 5 \\ 3 & 1 & -2 & 6 \end{pmatrix}$

2. Invertible square matrix. Show that a square matrix A is invertible if and only if its transpose A^T is invertible. Moreover, show that in that case we have $(A^T)^{-1} = (A^{-1})^T$.

3. Invertible matrices.

- (a) Let V, W be two n -dimensional vector spaces over a field K . Let $S \in \text{Hom}(V, W)$ and $T \in \text{Hom}(W, V)$ such that $T \circ S = \text{Id}_V$. Show that S is invertible and T is the inverse of S .
- (b) Let $A, B \in M_{n \times n}(K)$ such that $A \cdot B = I_n$. Show that A is invertible with inverse B .

4. Invertible similar matrices. Consider $n \times n$ -matrices A and B over K .

- (a) Show: If A or B is invertible, then AB and BA are similar.
- (b) Does this also hold true without the condition in (a)?

5. Gaussian elimination and invertibility. Determine with the help of Gaussian elimination, for which values of α the following matrix over $\mathbb{Q}$ is invertible:

$$\begin{pmatrix} 1 & 3 & -4 & 2 \\ 2 & 1 & -2 & 1 \\ 3 & -1 & -2 & -2\alpha \\ -6 & 2 & 1 & \alpha^2 \end{pmatrix}.$$

6. Bruhat decomposition. Prove the following:

Theorem. (*Bruhat-Decomposition*) For every invertible matrix A there exists a permutation matrix P , i.e. a matrix which has exactly one non-zero, which equals 1, in every column and every row, and invertible upper triangular matrices B and B' , such that

$$A = BPB'.$$

Hint: Choose an invertible upper triangular matrix U , such that the sum of the number of leading zeros in all rows of UA is maximal. Then find a permutation matrix Q , such that QUA is an invertible upper triangular matrix.

Multiplying a matrix A on the left by a permutation matrix P permutes the rows of A . For example, if P has a 1 at index (i, j) , the j -th line of A will become the i -th line of PA .

Single Choice. In each exercise, exactly one answer is correct.

1. Which assertion is not always satisfied?
 - (a) The base change matrix is the representation matrix of the identity map with respect to the according bases.
 - (b) Every finite dimensional vector space is isomorphic to K^n for some $n \geq 0$.
 - (c) The rank of a linear map $f : K^n \rightarrow K^m$ is at least $\min\{n, m\}$.
 - (d) The representation matrix of an isomorphism is invertible.
2. Consider $\mathbb{C}$ as real two-dimensional vectorspace with the ordered basis $\mathcal{B} := (1, i)$. The matrix $[\dots]_{\mathcal{B}}^{\mathcal{B}} := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is the representation matrix with respect to $\mathcal{B}$ of the linear map $\mathbb{C} \rightarrow \mathbb{C}$:
 - (a) Complex conjugation $z \mapsto \bar{z}$
 - (b) $z \mapsto \operatorname{Re}(z)$
 - (c) $z \mapsto \operatorname{Im}(z)$
 - (d) $z \mapsto iz$
3. The rank of $\begin{pmatrix} 1 & 3 \\ -3 & -9 \end{pmatrix}$ over $\mathbb{Q}$ is
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
4. For every $n \times m$ -Matrix A and every invertible $n \times n$ -matrix B , we have
 - (a) $\operatorname{rank}(BA) = \operatorname{rank}(B) \cdot \operatorname{rank}(A)$
 - (b) $\operatorname{rank}(BA) = \operatorname{rank}(B) + \operatorname{rank}(A)$
 - (c) $\operatorname{rank}(BA) = \operatorname{rank}(A)$
 - (d) $\operatorname{rank}(BA) = \operatorname{rank}(B)$

Multiple Choice Fragen.

1. Consider the following ordered bases of $\mathbb{R}[x]_2$:

$$\mathcal{B} = (1, x, x^2), \quad \mathcal{C} = (x^2, (x+1)^2, (x+2)^2)$$

Determine the base change matrix $Q := [\text{Id}_{\mathbb{R}[x]_2}]_{\mathcal{C}}^{\mathcal{B}}$.

(a) $Q = \begin{pmatrix} 0 & 1 & 4 \\ 0 & 2 & 4 \\ 1 & 1 & 1 \end{pmatrix}$

(b) $Q = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 1 & 4 \end{pmatrix}$

(c) $Q = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 2 & 1 \\ 4 & 4 & 1 \end{pmatrix}$

(d) $Q = \frac{1}{4} \begin{pmatrix} 2 & -4 & 4 \\ -1 & 4 & -3 \\ 4 & 0 & 0 \end{pmatrix}$

(e) $Q = \frac{1}{4} \begin{pmatrix} 2 & -3 & 4 \\ -4 & 4 & 0 \\ 2 & -1 & 0 \end{pmatrix}$

(f) $Q = \frac{1}{4} \begin{pmatrix} 2 & -1 & 4 \\ -4 & 4 & 0 \\ 4 & -3 & 0 \end{pmatrix}$

2. Let $A \in M_{n \times n}(K)$ and $P \in \text{GL}_n(K)$, denote

$$r := \text{rank}(A - I_n) \quad \text{and} \quad s := \text{rank}(PAP^{-1} - I_n).$$

Which of the following statements hold?

- (a) $r = s$;
- (b) $r \neq s$;
- (c) $r > s$;
- (d) $r < s$;
- (e) if $r < n$, $\exists v \in V \setminus \{0\}$ such that $Av = v$.